

Dosage Titration - Cell Culture

Introduction

Materials

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Procedure

Conditions (29)

1. Make Dox working solutions from stock of 50 mg/ml (~ 97 mM)

WD1: (1:500) 20 uL stock + 10 ml of media--> 100 µg/mL Dox

WD2: (1:5000) 1 mL WD1 + 9 ml of media--> 10 µg/mL Dox

WD3: (1:50,000) 1 mL WD2 + 9 ml of media--> 1 µg/mL Dox

2. Make TMP working solutions from stock of 100 mM

WD1: (1:100) 100 uL stock + 9.9 mL media --> 1 mM TMP

WD2: (1:1000) 1 mL WD1 + 9 mL media --> 100 uM TMP

WD3: (1:10,000) 1 mL WD2 + 9 mL media --> 10 uM TMP

Prep

3. Warm media

Make RPMI/puro media

4. Make drug working dilutions (WDs) - look above

Day 0 (Tuesday)

5. Get 8 12 well plates and 8 trenches

6. Count cells in:

CRISPRi GF11B (5) =

CRISPRi CD46 (11) =

CRISPRi NTC (3) =

CRISPRa GF11B (3) =

CRISPRa CD46 (6)=

K562 WT (1) =

7. Want 150,000 cells per well

Table2					
	A	B	C	D	E
1	Cell line	Amount of wells	Goal (cells) (M)	Volume (uL)	Resuspension Volume (mL)
2	CRISPRi GF11B	15	3		20
3	CRISPRi CD46	33	5.7		38
4	CRISPRi NTC	9	2.1		14
5	CRISPRa GF11B	9	2.1		14
6	CRISPRa CD46	18	3.45		23
7	K562 WT	3	1.2		8

8. Resuspend with specified volume of either puro/rpmi or normal rpmi for WT

9. Plate 2 mL from each condition (x3) in a 12 well plate

10. Make conditions by adding below amounts/drugs to each resuspended cell/media mix

Table1							
	A	B	C	D	E	F	G
1		Dox (ug/mL)	Dox (nM)	Dox (uL)	TMP (uM)	TMP (uL)	Total Volume (mL)
2	CRISPRi						
3	CD46 sgRNA						
4	Ctrl	0	0	0	0	0	2
5	D25	0.0123	25	24.6 WD3	5	10 WD1	2
6	D50	0.026	50	5.2 WD2	5	10 WD1	2
7	D200	0.10	200	2 WD1	5	10 WD1	2
8	D600	0.31	600	6.2 WD1	5	10 WD1	2
9	D2000	1.03	2000	20.6 WD1	5	10 WD1	2
10	T0.01	0.31	600	6.2 WD1	0.01	2 WD3	2
11	T0.1	0.31	600	6.2 WD1	0.1	2 WD2	2
12	T1	0.31	600	6.2 WD1	1	20 WD2	2
13	T5	0.31	600	6.2 WD1	5	10 WD1	2
14	T10	0.31	600	6.2 WD1	10	20 WD1	2
15	GFI1B sgRNA						
16	Ctrl	0	0	0	0	0	2
17	D50	0.026	50	5.2 WD2	5	10 WD1	2
18	D600	0.31	600	6.2 WD1	5	10 WD1	2
19	T0.01	0.31	600	6.2 WD1	0.01	2 WD3	2
20	T10	0.31	600	6.2 WD1	10	20 WD1	2
21	NTC sgRNA						
22	NTC baseline	0	0	0	0	0	2
23	tox ctrl 1	0.31	600	6.2 WD1	5	10 WD1	2
24	tox ctrl 2	1.03	2000	20.6 WD1	5	10 WD1	
25	CRISPRa						
26	CD46 sgRNA						
27	Ctrl	0	0	0	0	0	2
28	D25	0.0123	25	24.6 WD3	0	0	2
29	D50	0.026	50	5.2 WD2	0	0	2
30	D200	0.10	200	2 WD1	0	0	2
31	D600	0.31	600	6.2 WD1	0	0	2
32	D2000	1.03	2000	20.6 WD1	0	0	2
33	GFI1B sgRNA						
34	Ctrl	0	0	0	0	0	2
35	D50	0.026	50	5.2 WD2	0	0	2
36	D600	0.31	600	6.2 WD1	0	0	2
37	NTC sgRNA						
38							
39							
40							
41	WT K562						
42	WT baseline	0	0	0	0	0	2

11. Keep working solutions

Day 2 (Thursday) - split cells

12. Split cells by removing media/cells and leaving 0.5 mL and replacing it with 1.5 media and corresponding amounts of drug WDs

1:4 split

make media master mixes

Day 4 (Saturday)

13. Split cells by removing media/cells and leaving 0.5 mL and replacing it with 1.5 media and corresponding amounts of drug WDs

Day 6 (Tuesday)

14. Harvest!